

## MATH5320 – Assignment 5

### Question 1:

#### 1. Equivalence

In last week's homework, you computed the mean and variance of a data set for different window lengths as well as for exponential weighting with two different values for  $\lambda$ .

What  $\lambda$ s should be used for exponential weighting to get results most similar to using windows of 2, 5, and 10 years, respectively?

We estimated parameters the N-day Moving window of equal weights and an Exponential decaying weight, where, we have:

$$w_k = (1 - \lambda) \lambda^k, \text{ for } 0 < \lambda < 1, \quad k = 0, 1, 2, \dots$$

The average lag or the “centre of mass is:

$$\overline{k_{EWMA}} = \sum_{k=0}^{\infty} k w_k = (1 - \lambda) \sum_{k=0}^{\infty} k \lambda^k$$

And then, using the equation that approximates the sum of Ams:

$$\sum_{k=0}^{\infty} k \lambda^k = \frac{\lambda}{(1 - \lambda)^2},$$

$$\overline{k_{EWMA}} = \frac{\lambda}{(1 - \lambda)}$$

For an N-day window, the weights are  $1/N$  days on lags  $k$ , from  $0 \leq k \leq N - 1$ , therefore the mean lag will be:

$$\overline{k_{WIN}} = \frac{1}{N} \sum_{k=0}^{N-1} k = \frac{1}{N} \frac{(N-1)N}{2} = \frac{N-1}{2}$$

Then, we match the two averages such that:

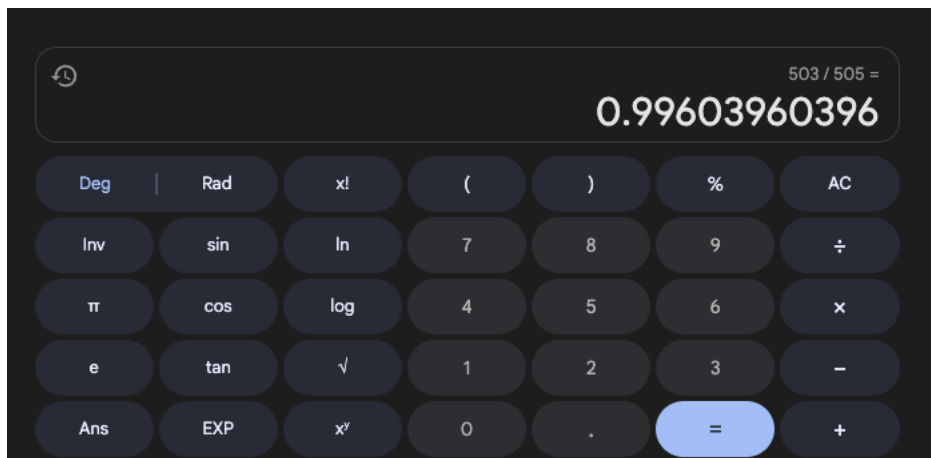
$$k = \frac{N-1}{2} = \frac{\lambda}{(1-\lambda)} \rightarrow \lambda = 1 - \frac{2}{N+1} = \frac{N-1}{N+1}$$

|   | A                     | B                       | C                               | D              |  |
|---|-----------------------|-------------------------|---------------------------------|----------------|--|
| 1 | Window Length (Years) | Window Length (Days, N) | Equivalent Lambda ( $\lambda$ ) | FormulaText    |  |
| 2 | 1y                    | 252                     | 0.992094862                     | =(B2-1)/(B2+1) |  |
| 3 | 2y                    | 504                     | 0.996039604                     | =(B3-1)/(B3+1) |  |
| 4 | 5y                    | 1260                    | 0.998413957                     | =(B4-1)/(B4+1) |  |
| 5 | 10y                   | 2520                    | 0.999206664                     | =(B5-1)/(B5+1) |  |
| 6 |                       |                         |                                 |                |  |
| 7 |                       |                         |                                 |                |  |

We get the following:

| Window Length (Years) | Window Length (Days, N) | Equivalent Lambda ( $\lambda$ ) |
|-----------------------|-------------------------|---------------------------------|
| 1y                    | 252                     | 0.992094862                     |
| 2y                    | 504                     | 0.996039604                     |
| 5y                    | 1260                    | 0.998413957                     |
| 10y                   | 2520                    | 0.999206664                     |

And we note that it matches:



## Question 2:

### 2. Historical estimates with equivalent lambda

Repeat last week's exponential weighting parameter estimation using the above computed lambdas equivalent to 2 year, 5 year and 10 year windows. Compare the windowed versions to the corresponding equivalent exponentially weighted versions.

We calculated the mean, variance, and volatility for both AAPL and CAT returns using the simple window method, and the exponentials:

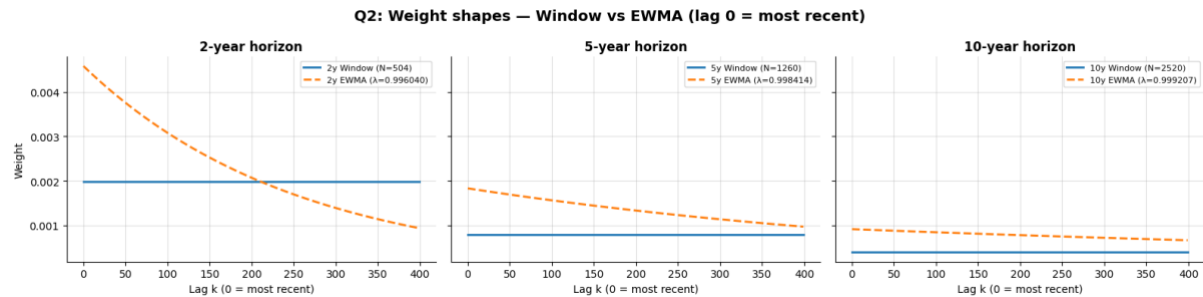
| AAPL - Window vs Exponential Weighting:                                                                                                                                          | CAT - Window vs Exponential Weighting:                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Year ( $\lambda=0.996040$ ):<br>Window: $\mu=0.000743$ , $\sigma^2=0.00030542$ , $\sigma=0.017476$<br>Exponential: $\mu=0.000889$ , $\sigma^2=0.00029353$ , $\sigma=0.017133$  | 2-Year ( $\lambda=0.996040$ ):<br>Window: $\mu=0.001733$ , $\sigma^2=0.00035983$ , $\sigma=0.018969$<br>Exponential: $\mu=0.002823$ , $\sigma^2=0.00040546$ , $\sigma=0.020136$  |
| 5-Year ( $\lambda=0.998414$ ):<br>Window: $\mu=0.000552$ , $\sigma^2=0.00029999$ , $\sigma=0.017320$<br>Exponential: $\mu=0.000678$ , $\sigma^2=0.00029558$ , $\sigma=0.017193$  | 5-Year ( $\lambda=0.998414$ ):<br>Window: $\mu=0.001109$ , $\sigma^2=0.00034630$ , $\sigma=0.018609$<br>Exponential: $\mu=0.001691$ , $\sigma^2=0.00036427$ , $\sigma=0.019086$  |
| 10-Year ( $\lambda=0.999207$ ):<br>Window: $\mu=0.000967$ , $\sigma^2=0.00032962$ , $\sigma=0.018155$<br>Exponential: $\mu=0.000864$ , $\sigma^2=0.00033009$ , $\sigma=0.018168$ | 10-Year ( $\lambda=0.999207$ ):<br>Window: $\mu=0.000994$ , $\sigma^2=0.00036839$ , $\sigma=0.019194$<br>Exponential: $\mu=0.001238$ , $\sigma^2=0.00037589$ , $\sigma=0.019388$ |

It is notable, that despite using the overhead *same mean*, the exponential and window estimates will diverge during regime shifts as EWMA re-weights recent data more aggressively, so that the outputs are more responsive, but potentially noisier. Note the example of CAT 2y, where window vol is about 0.019, whereas EWMA vol is 0.02, which is a noticeable jump. Noticeably, the differences are generally bigger at shorter horizons, and shrink for longer horizons, as shorter windows or faster decays, means higher sensitivity to recent regimes.

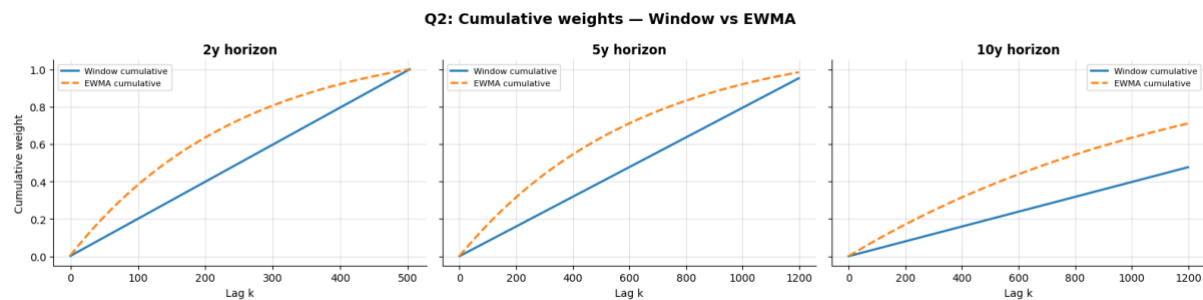
As seen in HW04, the lambda-weighted charts are more noisy, but more responsive, whereas the simple window gives equal weight to all data points within the window. The lambda-weight does give progressively less weight to older data points, like “memory”.

This showcases the stability and robustness issue, where small changes in market conditions should not produce wild swings in parameters, unless it is intended. In this case, even with the matched lambda, the window and EWMA aren't identical, as the weighing shapes differ. This is the definition of model risk, as downstream numbers for VaR/ES inherit these differences.

Looking further into the visualisations:



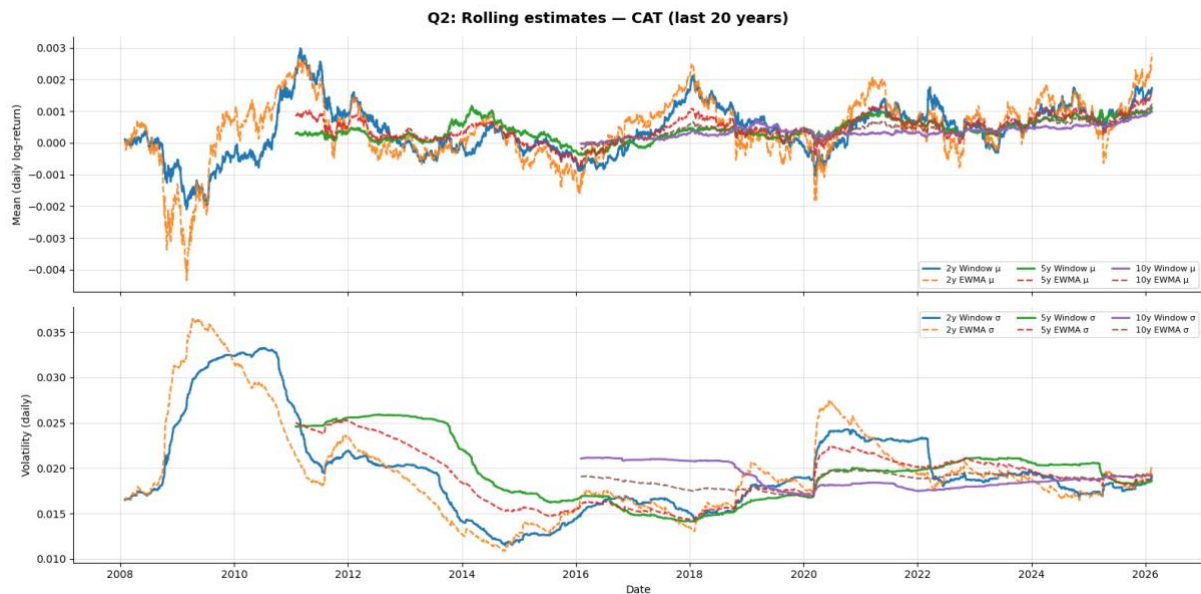
This figure illustrates the structural difference between the window estimator and the exponentially weighted moving average (EWMA), even when  $\lambda$  is calibrated to match the equivalent window length. The window assigns equal weight  $1/N$  to each observation within the horizon and zero weight beyond it, producing a flat weighting profile, whereas otoh, the EWMA is geometrically decaying. For longer horizons, both kernels are far more flat.



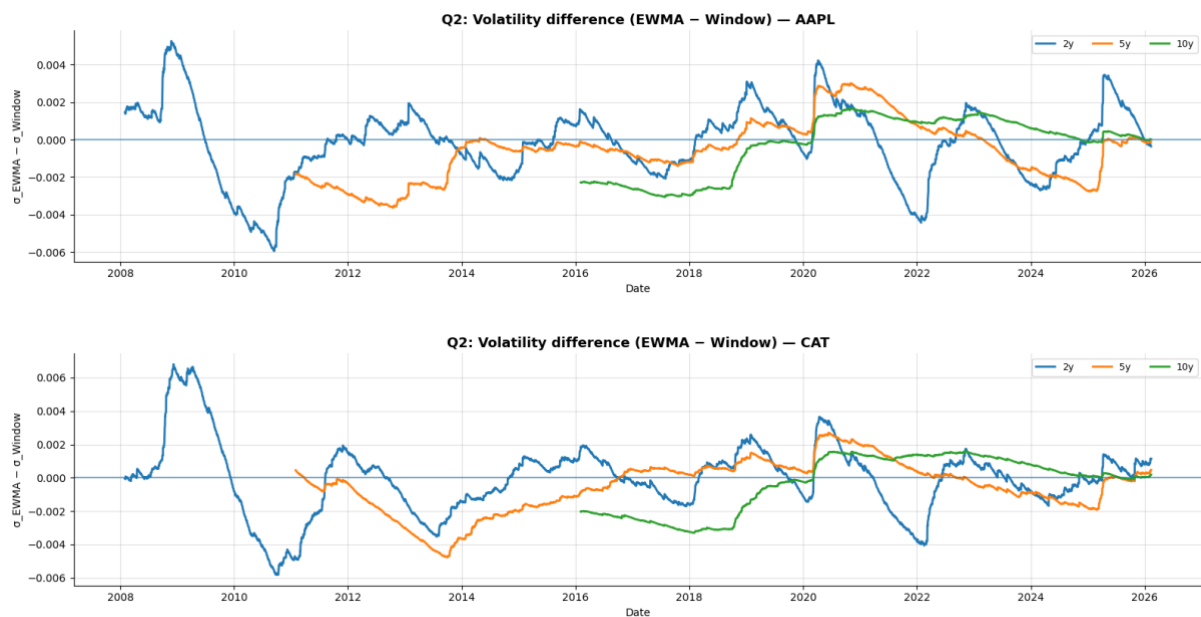
Here, for the cumulative weights, it provides the “Effective memory” interpretation o the estimators, and sjows that the window accumulates linearly, whereas the EWMA accumulates rapidly in the first segment, and then occurs slowly. It does, also show why EWMA tends to adapt faster during market stress.



Here, we see that the rolling mean plots compared across the two types, with largest divergence for the 2y. This is normal, as estimator choice matters more for shorter horizons.



Rolling vol, shows more pronounced differences, especially during periods of stress.



Vol, as a whole, shows pronounced differences, compared to the mean plots. During periods of elevated vol, EWMA rises faster, and it does appear that the window estimator remains elevated until high vol observations exit the rolling window.

Q2: Latest rolling estimates (full sample last date):

| Asset | Horizon | Window_mu | EWMA_mu  | Window_sigma | EWMA_sigma | RelDiff_sigma_% |
|-------|---------|-----------|----------|--------------|------------|-----------------|
| AAPL  | 2y      | 0.000743  | 0.000889 | 0.017494     | 0.017133   | -2.06           |
| AAPL  | 5y      | 0.000552  | 0.000678 | 0.017327     | 0.017193   | -0.78           |
| AAPL  | 10y     | 0.000967  | 0.000864 | 0.018159     | 0.018168   | 0.05            |
| CAT   | 2y      | 0.001733  | 0.002823 | 0.018988     | 0.020136   | 6.05            |
| CAT   | 5y      | 0.001109  | 0.001691 | 0.018617     | 0.019086   | 2.52            |
| CAT   | 10y     | 0.000994  | 0.001238 | 0.019197     | 0.019388   | 0.99            |

As we can see here, there is a large divergence in the lower levels of horizons, getting lesser as “time increases” for the time horizon.

### Question 3:

#### 3. Formula VaR and ES from historical estimates

Using the estimates for the drift and volatility from last week using 2 year, 5 year and 10 year windows, and using the corresponding equivalent exponential weights from above, tabulate and graph the  $\text{VaR}(S, T, p)$  and the  $\text{ES}(S, T, p')$  for  $S$  being  $A$  and  $C$ , using  $p = 0.99$ ,  $p' = 0.975$ , and  $T = 5$  days (i.e. one week), for each day over the last 20 years, or as far back as the above estimates were done. Tabulate and graph.

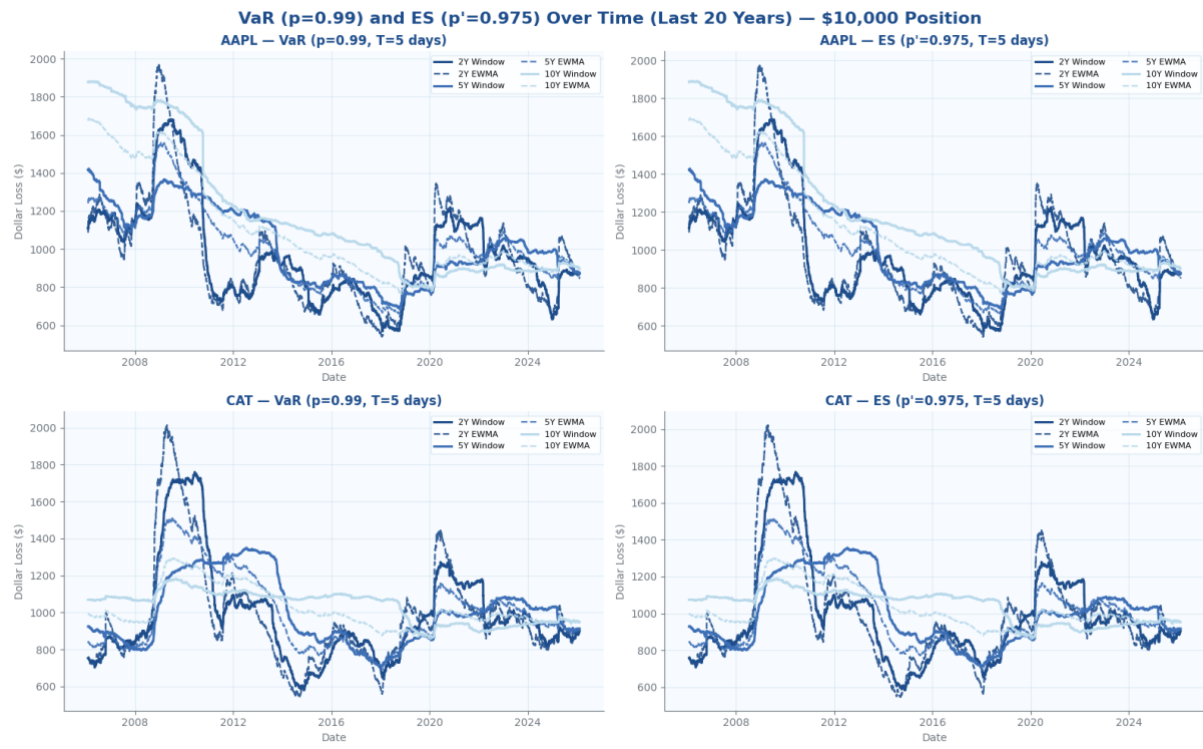
Unlike the parameter estimates, the computed VaRs and ESs will not be comparable between dates if you compute them on 1 share of stock. They can be normalized by either computing them assuming a \$10,000 position each day, or converting them to relative losses. Do this exercise with the former normalization.

We are asked to compute 5-day risk measure (1 trading week), by looking at the 0.99 VaR & 0.975 ES, with a normalised \$10,000 position each day, such that the results are comparable over time.

In this case, we estimate the drift and vol, through the windowed lookbacks for 2y/5y/10y, and then looking at the corresponding weights above.

Like before, the daily log returns were used to model, and assuming iid returns, we have the scale the mean return by  $T$ , and the volatility by  $\sqrt{T}$ , as is custom.

Notably, we have a Loss, such that it is normal, as a % based of the \$10,000 given. Then we have the VaR, and ES, which is based off the CDF, at the corresponding  $p$  values. Note that ES should always be larger than the VaR.



Note: The above looks *extremely similar*, so minor sanity checks was made, such that we have:

```
print("Ratio ES/VaR (latest AAPL 2Y window):",
      aapl_q3['ES_window_2y'].iloc[-1] /
      aapl_q3['VaR_window_2y'].iloc[-1])
[56] ✓ 0.0s
... Ratio ES/VaR (latest AAPL 2Y window): 1.0051336847768444
```

| Time-average VaR/ES over the last 20y (NaNs skipped): |     |         |         |        |        |         |         |
|-------------------------------------------------------|-----|---------|---------|--------|--------|---------|---------|
| Horizon                                               |     | 10y     |         | 2y     |        | 5y      |         |
| Method                                                |     | exp     | window  | exp    | window | exp     | window  |
| Asset Measure                                         |     |         |         |        |        |         |         |
| AAPL                                                  | ES  | 1103.88 | 1187.36 | 971.02 | 978.06 | 1003.98 | 1023.33 |
|                                                       | VaR | 1098.20 | 1181.28 | 966.01 | 973.01 | 998.79  | 1018.03 |
| CAT                                                   | ES  | 1025.61 | 1039.73 | 987.14 | 989.36 | 1002.08 | 1005.33 |
|                                                       | VaR | 1020.49 | 1034.54 | 982.20 | 984.41 | 997.07  | 1000.30 |

Notably, we can see that like before, shorter window lengths, result in more volatility risk estimates that react far more quickly to changing market conditions. We can see that 2y is most reactive, and 10y the smoothest. We note that ES is always above VaR at the same time, with a more conservative tail average. In the plots, that can be seen, and that is to be expected.

Furthermore, it is evident that the EWMA reacts slightly earlier than the same length window around regime shifts, given that it places a more heavy weight on recent returns, and the dashed vs solid lines above showcases these differences.

We can see that in periods of stress like 2008-09, and COVID, there was a large spike in both VaR and ES, especially CAT in the bottom columns, being extremely volatile driven.

The equivalent lambda does show that across the horizon, even if it is matched, the weighting shape differs, so the risk estimates differ. We can see that the EWMA lines are usually close to the window lines in level, but the EWMA responds a bit faster when vol rises or falls. Differences are generally most visible around the turning points in recovery or crisis. This does demonstrate model risk to an extent, because despite two “same” estimators, both are on different paths.

#### Question 4:

##### 4. Derivative errors for normal CDF

Let  $F(x)$  be a function, and let  $f(x)$  be its derivative. Define the difference derivative of  $F$  with a step size  $h$  as

$$D(F, x, h) = (F(x + h) - F(x - h)) / (2h)$$

and then

$$D(F, x, h) \approx f(x)$$

We often use the above to approximate  $f$ . The error at  $x$  is then:

$$DErr(F, f, x, h) = f(x) - D(F, x, h)$$

The graph of  $DErr(F, f, x, h)$  or even of  $D(F, x, h)$  can be used to choose an optimal  $h$  for computing the derivative and to illustrate the accuracy of  $D(F, x, h)$ .

Let  $\phi(x)$  and  $\Phi(x)$  be the standard normal distribution, and CDF, respectively. Then

$$\phi(x) = d\Phi(x)/dx$$

In lecture 5, a graph was displayed to illustrate the error  $DErr(\Phi, \phi, h)$  for different values of  $h$ . Reproduce this graph using your own software to estimate the optimal  $h$  to use in computing the derivative of  $\Phi$ . Also graph  $\phi(x)$  and  $D(\Phi, x, h)$  on the same plot to see if the errors are visible directly on the plots.

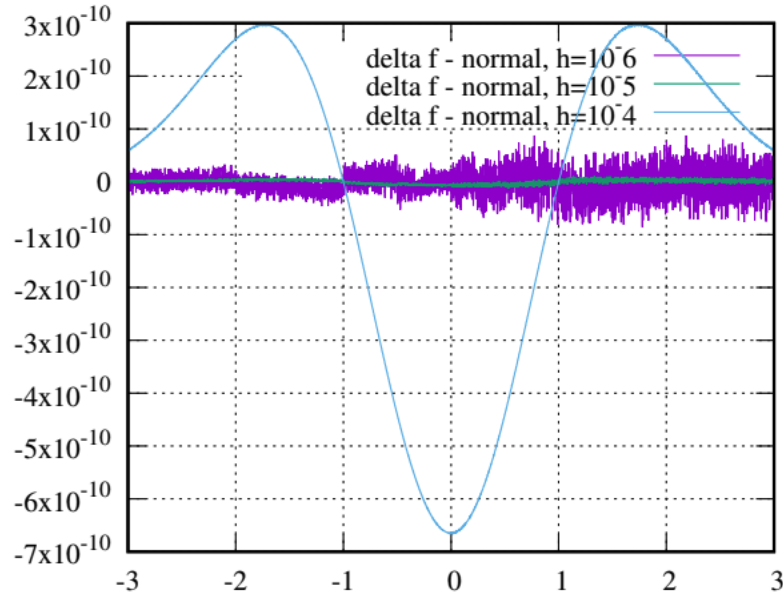
In this question, we are approximating the derivative of a function, using the central difference method. Then, we compare it to the true derivative via the derivative error. We intend to replicate the image below from slide 49 from the lecture:



## Analytic errors

Even analytic calculations are subject to such errors.

Know derivative of Normal CDF, so compare to difference derivative:



The error was defined as:

$$DErr(\Phi, \phi, x, h) = \phi(x) - D(\Phi, x, h)$$

We plot the **DErr**, versus  $x$  within  $[-3, 3]$ , for  $h$  for  $10^{-6}, 10^{-5}, 10^{-4}$ , for larger  $h$ , that the error curve is smoothly and highly structured, and we can see that it is dominated by the truncation error, from the central-difference approximation, considering the Taylor expansion around  $x$ , we have:

$$\Phi(x + h) = \Phi(x) + h\phi(x) + \frac{h^2}{2}\phi'(x) + \frac{h^3}{6}\phi''(x) + O(h^4),$$

$$\Phi(x - h) = \Phi(x) - h\phi(x) + \frac{h^2}{2}\phi'(x) - \frac{h^3}{6}\phi''(x) + O(h^4)$$

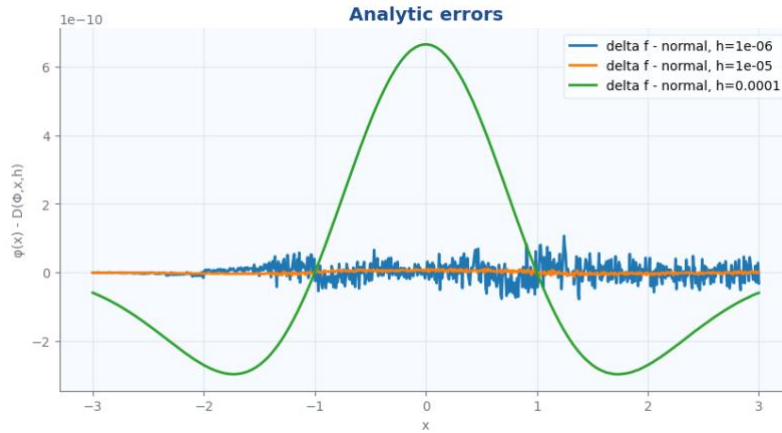
If we consider both here, subtracting it, and then dividing by  $2h$ , gives the central difference, shown as follow:

$$D(\Phi, x, h) = \frac{\Phi(x + h) - \Phi(x - h)}{2h} = \phi(x) + \frac{h^2}{6}\phi''(x) + O(h^4) \text{ (since } \Phi'(x) = \phi(x) \text{)}$$

Then, the derivative error is:

$$DErr(\Phi, \phi, x, h) = \phi(x) - D(\Phi, x, h) = -\frac{h^2}{6}\phi''(x) + O(h^4),$$

As such, we can see that the leading-order error, will scale like  $h^2$ , will explain why  $h = 10^{-4}$ , produces a visibly larger error curve than the other values, and shows why theoretically the other values should provide a smaller truncation error. We can see that the large  $h$  for  $10^{-4}$ , the error is smooth and shows the truncation error from the Taylor approximation. On the other hand, small  $h$ , it has much lower errors. But for  $10^{-6}$ , we can see that the truncation error is minimised, and we start to see floating point roundoff or cancellation.



Looking into this error, where we compare the truncation vs roundoff, and we use the Taylor Expansion:

$$\Phi(x + h) = \Phi(x) + \phi(x)h + \frac{\Phi^3(x)}{h^3} + \dots$$

$$\Phi(x - h) = \Phi(x) - \phi(x)h + \frac{\Phi^3(x)}{h^3} + \dots$$

And then, we subtract and divide by  $2h$  gives:

$$D(\Phi, x, h) = \phi(x) + \frac{\Phi^3(x)}{6}h^2 + O(h^4),$$

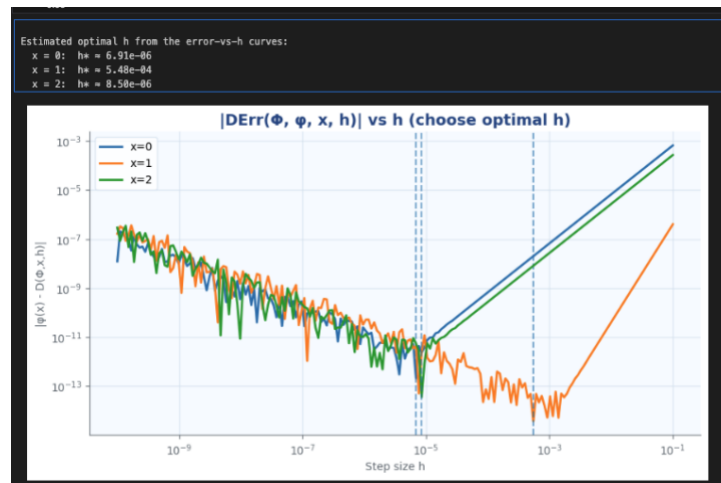
So then, here. we can see that truncation error scales pretty similar to  $O(h^2)$ :

$$\phi(x) - D(\Phi, x, h) = \frac{-\Phi^3(x)}{6}h^2$$

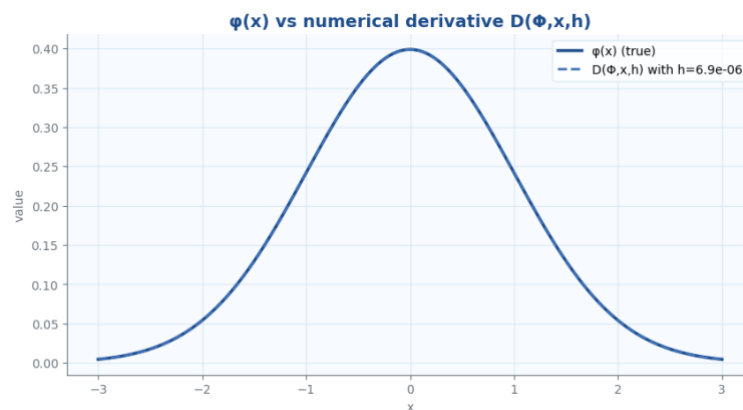
And the idea is that when  $h$  is extremely small, the subtraction of  $\Phi(x + h) - \Phi(x - h)$ , suffers from cancellation and floating-point noise. A simple model can be considered something like:

Truncation error (scales with  $h^2$ ), and the roundoff error, which is about  $\epsilon/h$ , which is scaled by machine epsilon. By inspection, we can estimate the  $h^*$  from about  $10^{-5}$  to  $10^{-6}$ . And then, as shown below the best choice is about:

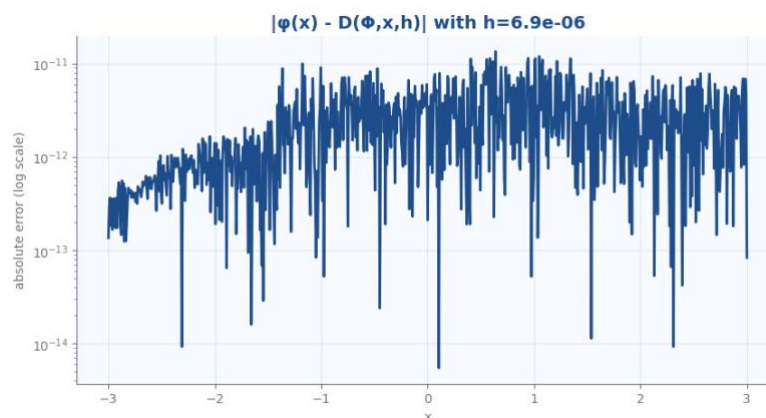
$$h^* = 6.9 * 10^{-6}$$



Then, we then overlaid the two plots together, they are visually almost identical, as such the error is too small to see at the function itself. Showing below, this is evident:



An alternative approach was taken, below where we took the log-difference, and shows that the remaining error  $10^{-12}$  to  $10^{-10}$  across most of  $[-3, 3]$ , which is pretty consistent with the earlier plots:



As such, it can be shown that the second-order accurate derivative approximation, but this precision creates a trade-off, where a decreasing  $h$  will reduce the truncation error, until the round-off error dominates. The best step size was found to be  $h^* = 6.9 \times 10^{-6}$ .

## Question 5:

### 5. Black Scholes delta error

Repeat the previous problem using the Black-Scholes formula for the price of a call option maturing in 2.0 years with a strike of 85.0. Use an implied volatility of 30.0 and a risk free rate of 4.5 % to look at errors as a function of starting stock price.

Plot the BS price and the delta itself as well to make sure your calculations make sense.

---

Noting the BSM model with the following parameters:

```
def difference_derivative(F, x, h):
    return (F(x + h) - F(x - h)) / (2.0 * h)

def black_scholes_call(S, K, T, r, sigma):
    S = float(S)
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S * stats.norm.cdf(d1) - K * np.exp(-r * T) * stats.norm.cdf(d2)

def black_scholes_delta(S, K, T, r, sigma):
    S = float(S)
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    return stats.norm.cdf(d1)
```

✓ 0.0s

K = 85.0  
T = 2.0  
r = 0.045  
sigma = 0.30

✓ 0.0s

We compute  $C(S_t, t)$ , which is given by:

$$C(s) = S\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

And then, we computed:

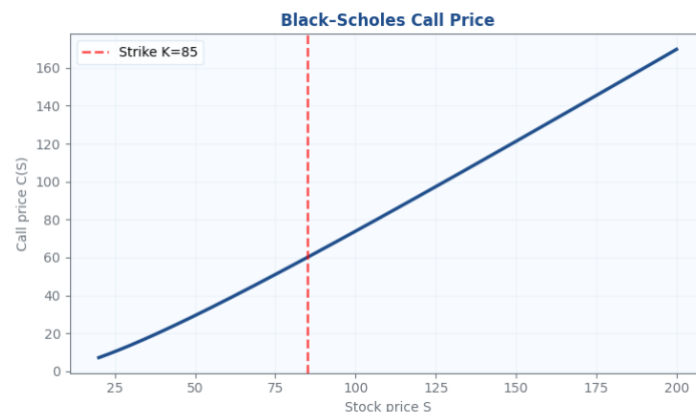
$\Delta_{analytic}(S) = \frac{\partial S}{\partial C} = \Phi(d_1)$ , so then we run the same central difference operator as before:

$$\Delta_C D(S; h) = D(C, S, h) = \frac{C(S + h) - C(S - h)}{2h}$$

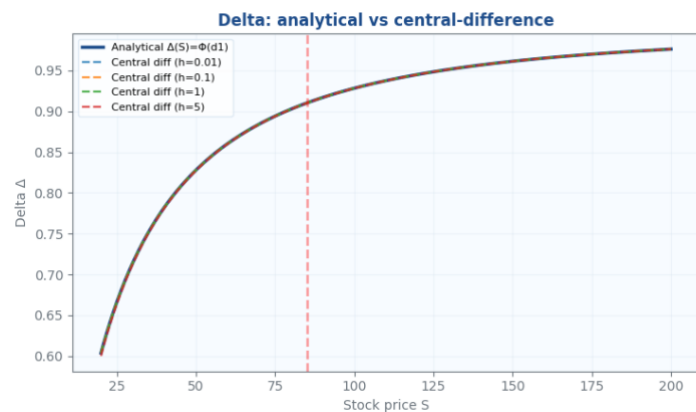
Following this we define, and then plot the delta-error as:

$$Err(S; h) = |\Delta_{analytic}(S) - \Delta_{CD}(S; h)|$$

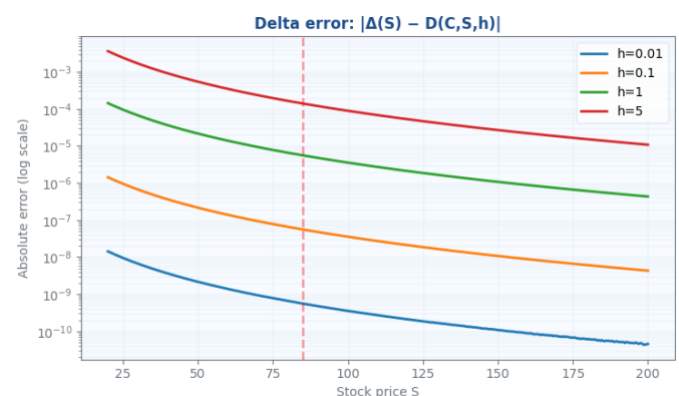
In this case, we consider the BSM price, and as shown, the curve is increase and convex as a call price should be. As shown, the vertical dashed line is at  $K = 85$ , which is the most visible.



The analytic delta is smooth, and naturally increases with  $S$ . Visibly, all the different  $h$  sits basically on top of the analytic curve. In this case, we can see that the  $C(S)$ , is smooth, and the central difference is second-order accurate when  $h$  is reasonable



Then, we can see that finally, the model risk part is predicated in the delta error. In this case, we can see that at  $h=5$ , the error is the largest because of truncation error, and the finite difference is far too coarse, and we are using points that are too far apart. For the moderately smaller  $h$ , like  $h=1$ ,  $h=0.1$ , improves substantively, where we are truer to the true derivative. For blue, the smallest  $h=0.01$ , it gives the smallest error in this range, and in this case, the error generally decreases as  $S$  increases, and as  $S$  increases relative to  $K$ , the call becomes deeply ITM,  $C(S)$  becomes linear in  $S$ , and the delta becomes close to 1 (as shown in the first plot), so it is far easier to approximate accurately.



As such, we replicated the central-difference idea for the BSM model, and we can see that it is broadly validated. The price and delta plots, does validate the implementation, with a monotone price, and strong changes around  $K$ . Noticeably, the error plot, shows the classic numerical trade-off, with large  $h$  leading to truncation errors, and vice versa. In this case,  $h = 0.01$ , produces the lowest delta error across the  $S$  range.